

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, JavaScript/TypeScript, C/C++, Java, SQL

Tools and Frameworks: PyTorch, LangChain, CrewAI, Qdrant, Django, FastAPI, React, Next.js, MongoDB, PostgreSQL, PyTest, GoogleTest, AWS, Docker, Kubernetes, GitHub Actions(CI/CD), Claude Code

Areas of Interest and Coursework: AI & LLMs, Data Structures & Algorithms, Software Engineering (SDLC), Distributed System Design, Object Oriented Design (OOD), Computer Architecture

INDUSTRY EXPERIENCE

Data AI Intern | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Developed backend services for an AI-native GRC platform automating NIST 800-53 compliance workflows for **~1,200** controls for RFP bidders, cutting reviews from **days of manual work to minutes**.
- Implemented a fallback mechanism across 5 embedding models (OpenAI, Ollama) that re-runs RAG queries with insufficient control citation hits, lifting retrieval from **32/50 to 50/50** hand-reviewed compliance queries.
- Diagnosed an OOM bug where freshly pulled 4–5 GB LLM weights lingered in the OS page cache; fixed it with targeted `posix_fadvise` eviction, **eliminating model-load failures**.

Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Built a React + Django AI assignment-review platform to flag rubric mistakes, cutting coach prep from **5 to 2 hours**.
- Architected a fully serverless AWS stack using an SQS-triggered Lambda AI worker, DynamoDB, and S3/CloudFront, sustaining 200+ submissions across 5 cohorts at **~\$3/month** idle cost.

Full Stack Developer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Engineered an admin-configurable cost estimation engine, cutting quote turnaround from **~2 days to <1 min**.
- Implemented JWT auth and idempotent Stripe webhooks, moving deposit collection online with **zero** duplicate charges.
- Sustained **300 req/s** at **<100 ms** p95 latency on one EC2 instance via Redis caching and PostgreSQL indexing.

ACADEMIC EXPERIENCE

Graduate Assistant | UMBC - Computer Architecture & Software Engineering

Aug 2025 – June 2026

- Designed a C++ assignment on virtual memory (testing knowledge of MMU address translation, multi-level page tables, and kernel page-fault handling) and graded coursework in computer architecture for **40+** students.
- Mentored 10 student **Scrum** teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and **stakeholders**, keeping all 10 projects on schedule amid evolving requirements.

Teaching Assistant | UMBC - Data Structures

Jan 2025 – June 2025

- Developed 20+ unit tests for the hash table project, automating logic checking and reducing grading time by **80%**.
- Helped **50+** students design and debug C++ projects implementing core data structures and graded homework.

PROJECTS

Bloodborne TAG | Text Adventure Game with natural language command parsing

- Engineered a command parser using few-shot Mistral-7B for slot extraction and embedding-similarity-based intent matching, resolving every free-form playtest command to one of **14 deterministic actions**.
- Architected a data-driven game engine with a **41-room** world and 60+ items/NPCs, extensible without code changes.

When Pixels Talk Back | Watermarking EMR Data to analyse impact on CV models

- Embedded a 1024-bit blind fragile watermark into RONI regions of 2,482 COVID-19 lung CT scans, achieving near-lossless quality (SSIM **0.9999**) while preserving DenseNet-169 performance (F1 **0.952**).
- Benchmarked 3 watermarking algorithms against 3 Computer Vision models, showing robust watermarks cut positive-case detection to **48%** and skew errors toward false negatives.

Nothing Lost in Translation | Lexically constrained Sequence Generation

- Built a Grid Beam Search decoder guaranteeing positive/negative lexical constraint satisfaction and raising BLEU by **15 points** (25.45 → 40.59) for English-to-Russian NMT.
- Reconstructed a beam search decoder from scratch, outperforming HuggingFace's stock `.generate()` by **1.45 BLEU**.